

AI ASSISTED CODING ASSIGNMENT - 3.1

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Task Description

Design and refine prompts using different prompting strategies to generate Python programs for basic computational problems.

Question 1: Zero-Shot Prompting (Palindrome Number Program)

Write a **zero-shot** prompt (without providing any examples) to generate a Python function that checks whether a given number is a palindrome.

• Task Description

- Record the AI-generated code.
- Test the code with multiple inputs.
- Identify any logical errors or missing edge-case handling.

```
areal.py > ...
1  # display a Python program that checks whether a given number is a palindrome
2  def is_palindrome(n):
3      if not isinstance(n, int):
4          return "Invalid number"
5      elif n < 0:
6          return "Not a palindrome"
7      else:
8          original = n
9          reversed_num = 0
10         while n > 0:
11             digit = n % 10
12             reversed_num = reversed_num * 10 + digit
13             n = n // 10
14         if original == reversed_num:
15             return f"{original} is a palindrome"
16         else:
17             return f"{original} is not a palindrome"
18 n=int(input("Enter a number: "))
19 print(is_palindrome(n))
20
```

Question 2: One-Shot Prompting (Factorial Calculation)

Write a one-shot prompt by providing one input-output example and ask the AI to generate a Python function to compute the factorial of a given number.

• Example:

- Input: 5 → Output: 120

• Task Description

- Compare the generated code with a zero-shot solution.
- Examine improvements in clarity and correctness.

```
#n = 5
#print factorial of n is 120
def factorial(n):
    if n == 0:
        return 1
    else:
        return n * factorial(n-1)
num = int(input("Enter a number: "))
print("Factorial of", num, "is", factorial(num))
```

Question 3: Few-Shot Prompting (Armstrong Number Check)

Write a few-shot prompt by providing multiple input-output examples to guide the AI in generating a Python function to check whether a given number is an Armstrong number.

- **Example:**

- Input: 153 → Output: Armstrong Number
- Input: 370 → Output: Armstrong Number
- Input: 123 → Output: Not an Armstrong Number

- **Task Description**

- Analyse how multiple examples influence code structure and accuracy.
- Test the function with boundary values and invalid inputs.

```
assain3.py > ...
47
48 n = 153
49 display Armstrong number
50 n = 370
51 display Armstrong number
52 n = 121
53 display not Armstrong number
54 ...
55
56 def is_armstrong(n):
57     if not isinstance(n, int):
58         return "Invalid number"
59     elif n < 0:
60         return "Not an Armstrong number"
61     else:
62         num_str = str(n)
63         num_digits = len(num_str)
64         sum_of_powers = sum(int(digit) ** num_digits for digit in num_str)
65         if sum_of_powers == n:
66             return f"{n} is an Armstrong number"
67         else:
68             return f"{n} is not an Armstrong number"
69
70 try:
71     n = int(input("Enter a number: "))
72     print(is_armstrong(n))
73 except ValueError:
74     print("Invalid number")
```

Question 4: Context-Managed Prompting (Optimized Number Classification)

Design a context-managed prompt with clear instructions and constraints to generate an optimized Python program that classifies a number as prime, composite, or neither.

• Task Description

- Ensure proper input validation.
- Optimize the logic for efficiency.
- Compare the output with earlier prompting strategies.

```

reading input a number and
check the number is prime or composite or neither'''

def check_prime_composite(n):
    if not isinstance(n, int):
        return "Invalid number"
    elif n < 0:
        return "Invalid number"
    elif n == 0 or n == 1:
        return f"{n} is neither prime nor composite"
    else:
        for i in range(2, int(n**0.5) + 1):
            if n % i == 0:
                return f"{n} is a composite number"
        return f"{n} is a prime number"

try:
    n = int(input("Enter a number: "))
    print(check_prime_composite(n))
except ValueError:
    print("Invalid number")

```

Question 5: Zero-Shot Prompting (Perfect Number Check)

Write a zero-shot prompt (without providing any examples) to generate a Python function that checks whether a given number is a perfect number.

• Task Description

- Record the AI-generated code.
- Test the program with multiple inputs.
- Identify any missing conditions or inefficiencies in the logic.

```

#display a Python program that checks whether a given number is a perfect number

def is_perfect_number(n):
    if not isinstance(n, int):
        return "Invalid number"
    elif n < 1:
        return "Not a perfect number"
    else:
        divisors_sum = sum(i for i in range(1, n) if n % i == 0)
        if divisors_sum == n:
            return f"{n} is a perfect number"
        else:
            return f"{n} is not a perfect number"
try:
    n = int(input("Enter a number: "))
    print(is_perfect_number(n))
except ValueError:
    print("Invalid number")

```

Question 6: Few-Shot Prompting (Even or Odd Classification with Validation)

Write a few-shot prompt by providing multiple input-output examples to guide the AI in generating a Python program that determines whether a given number is even or odd, including proper input validation.

- **Example:**

- Input: 8 → Output: Even
- Input: 15 → Output: Odd
- Input: 0 → Output: Even

- **Task Description**

- Analyse how examples improve input handling and output clarity.
- Test the program with negative numbers and non-integer inputs.

```

...
num = 28
display_even_number
num = 7
display_odd_number
num = 5
display_invalid_number
num = sfsd
display_invalid_number
...

def check_odd_even_extended(number):
    if not isinstance(number, int):
        return "Invalid number"
    elif number < 0:
        return "Invalid number"
    elif number % 2 == 0:
        return f"[number] is an even number"
    else:
        return f"[number] is an odd number"

try:
    number = int(input("Enter a number: "))
    print(check_odd_even_extended(number))
except ValueError:
    print("Invalid number")

```